

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication	20250269532
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	TAKUSHIMA; Takayoshi et al.

SUBSTRATE TRANSPORT SYSTEM

Abstract

The system has: a transport section that transports a substrate so that the substrate reaches A position, B position, C position, the B position, and the A position in this order, a A sensor that detects presence of the substrate that has reached the A position, a B sensor that detects presence of the substrate that has reached the B position, a timekeeping section that measures elapsed time when the A sensor and the B sensor detect the presence of the substrate in this order, and an abnormality detection unit that photographs the substrate that has reached the C position, and detects presence or absence of an abnormality in transport of the substrate based on the elapsed time measured by the timekeeping section and an image obtained by the photographing.

Inventors:	TAKUSHIMA; Takayoshi (Tokyo, JP), MARUYAMA; Kento (Tokyo, JP)
Applicant:	SUMCO Corporation (Tokyo, JP)
Family ID:	1000008407013
Assignee:	SUMCO Corporation (Tokyo, JP)
Appl. No.:	19/008977
Filed:	January 03, 2025

Foreign Application Priority Data

JP	2024-025640	Feb. 22, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: B25J9/16 (20060101); B25J11/00 (20060101)

U.S. Cl.:

CPC B25J9/1674 (20130101); B25J9/1697 (20130101); B25J11/0095 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a substrate transport system.

BACKGROUND

[0002] A substrate transport system for transporting substrates has been known (see, for example, Patent Document 1).

CITATION LIST

Patent Literature

[0003] PTL 1: JP 2013-110200 A1

SUMMARY

[0004] The object of this disclosure is to provide a substrate transport system that can reduce the burden of data processing when detecting, by photographing, the presence or absence of an abnormality in transport of a substrate.

[0005] One aspect of the present disclosure is as follows. [0006] [1] A substrate transport system having: [0007] a transport section that transports a substrate so that the substrate reaches A position, B position, C position, the B position, and the A position in this order, [0008] a A sensor that detects presence of the substrate that has reached the A position, [0009] a B sensor that detects presence of the substrate that has reached the B position, [0010] a timekeeping section that measures elapsed time when the A sensor and the B sensor detect the presence of the substrate in this order, and [0011] an abnormality detection unit that photographs the substrate that has reached the C position, and detects presence or absence of an abnormality in transport of the substrate based on the elapsed time measured by the timekeeping section and an image obtained by the photographing. [0012] [2] The substrate transport system as described in [1], wherein the abnormality detection unit performs the photographing at a timing determined based on the elapsed time measured by the timekeeping section. [0013] [3] The substrate transport system as described in [1] or [2], having a notification section that makes a notification when the abnormality detection unit detects the presence of an abnormality in transport of the substrate. [0014] [4] The substrate transport system as described in [3], having: [0015] a second transport section that transports a second substrate so that the second substrate reaches second A position, second B position, second C position, the second B position, and the second A position in this order, [0016] a second A sensor that detects presence of the second substrate that has reached the second A position, and [0017] a second B sensor that detects presence of the second substrate that has reached the second B position, wherein [0018] the timekeeping section measures second elapsed time when the second A sensor and the second B sensor detect the presence of the second substrate in this order, [0019] the abnormality detection unit performs second photographing of the second substrate that has reached the second C position, and detects presence or absence of an abnormality in transport of the second substrate based on the second elapsed time measured by the timekeeping section and an image obtained by the second photographing, and [0020] the notification section makes a notification when the abnormality detection unit detects the presence of an abnormality in transport of the second substrate.

[0021] According to the present disclosure, it is possible to provide a substrate transport system that can reduce the burden of data processing when detecting, by photographing, the presence or absence of an abnormality in transport of a substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] In the accompanying drawings:

[0023] FIG. 1 is a top view of a first transport processing unit of the substrate transport system in accordance with one embodiment of the present disclosure;

[0024] FIG. 2 is an A-A cross-sectional view of FIG. 1, illustrating the state when the first substrate being transported by the first transport section has reached first A position;

[0025] FIG. 3 is a cross-sectional view illustrating the state when the first substrate has reached first B position after being transported from the state illustrated in FIG. 2;

[0026] FIG. 4 is a cross-sectional view illustrating the state when the first substrate has reached first C position after being transported from the state illustrated in FIG. 3; and

[0027] FIG. 5 is a schematic diagram illustrating the overall substrate transport system illustrated in FIG. 1.

DETAILED DESCRIPTION

[0028] Hereinafter, an embodiment of the present disclosure will be described with reference to the drawings.

[0029] As illustrated in FIGS. 1 to 5, in one embodiment of the present disclosure, the substrate transport system **1** (see FIG. 5) has:

[0030] a (first) transport section **3A** (see FIG. 1) that transports a substrate **11A** so that the substrate **11A** reaches (first) A position **PIA**, (first) B position **P2A**, (first) C position **P3A**, the B position **P2A**, and the A position **PIA** in this order (see FIGS. 2 to 4), a (first) A sensor **12A** (see FIG. 1) that detects presence of the substrate **11A** that has reached the A position **PIA**, a (first) B sensor **13A** (see FIG. 1) that detects presence of the substrate **11A** that has reached the B position **P2A**, a timekeeping section **14** (see FIG. 5) that measures (first) elapsed time when the A sensor **12A** and the B sensor **13A** detect the presence of the substrate **11A** in this order, and an abnormality detection unit **7** (see FIG. 5) that photographs the substrate **11A** that has reached the C position **P3A**, and detects presence or absence of an abnormality in transport of the substrate **11A** at the C position **P3A** based on the elapsed time measured by the timekeeping section **14** and an image obtained by the photographing.

[0031] As illustrated in FIG. 5, in this embodiment, the substrate transport system **1** has a (first) transport processing unit **2A** which has the (first) transport section **3A**, a (first) transport pallet **4A**, and a (first) processing section **5A**; and a second transport processing unit **2B** which has a second transport section **3B**, a second transport pallet **4B**, and a second processing section **5B**. The (first) transport pallet **4A** has a (first) gripper **4aA** as a (first) system component **6A** and a (first) pallet **4bA**. The second transport pallet **4B** has a second gripper **4aB** as a second system component **6B** and a second pallet **4bB**. In addition, the substrate transport system **1** has a third transport processing unit **2C**, a fourth transport processing unit **2D**, a fifth transport processing unit **2E**, and a sixth transport processing unit **2F**. It will be noted that the number of the transport processing units in the substrate transport system **1** can be set as appropriate. The (first) transport processing unit **2A** transports and processes the (first) substrate **11A**, the second transport processing unit **2B** transports and processes the second substrate **11B**, the third transport processing unit **2C** transports and processes the third substrate **11C**, the fourth transport and processing unit **2D** transports and processes the fourth substrate **11D**, the fifth transport processing unit **2E** transports and processes the fifth substrate **11E**, and the sixth transport processing unit **2F** transports and processes the sixth substrate **11F**.

[0032] The substrate transport system **1** has the abnormality detection unit **7** (see FIG. 5) which has a (first) imaging section **7aA** for the (first) transport processing unit **2A**, a second imaging section **7aB** for the second transport processing unit **2B**, and a judgment section **7b**; and a notification section **9** (see FIG. 5). The judgment section **7b** is capable of communicating with the notification section **9**, the (first) imaging section **7aA**, and the second imaging section **7aB**. In addition, the

abnormality detection unit **7** has a third imaging section **7aC** for the third transport processing unit **2C**, a fourth imaging section **7aD** for the fourth transport processing unit **2D**, a fifth imaging section **7aE** for the fifth transport processing unit **2E**, and a sixth imaging section **7aF** for the sixth transport processing unit **2F**.

[0033] The substrate transport system **1** has the (first) A sensor **12A** and the (first) B sensor **13A** for the (first) transport processing unit **2A**, and the (second) A sensor **12B** and the (second) B sensor **13B** for the (second) transport processing unit **2B**. The timekeeping section **14** is capable of communicating with the (first) A sensor **12A**, the (first) B sensor **13A**, the second A sensor **12B**, the second B sensor **13B**, and the abnormality detection unit **7**. In addition, the substrate transport system **1** has a third A sensor **12C** and a third B sensor **13C** for the third transport processing unit **2C**, a fourth A sensor **12D** and a fourth B sensor **13D** for the fourth transport processing unit **2D**, a fifth A sensor **12E** and a fifth B sensor **13E** for the fifth transport processing unit **2E**, and a sixth A sensor **12F** and a sixth B sensor **13F** for the sixth transport processing unit **2F**.

[0034] In this embodiment, when the substrate **11A** (such as a silicon wafer) is transported by the transport section **3A** to reach the A position **P1A**, the B position **P2A**, and the C position **P3A** in this order, the abnormality detection unit **7** detects the presence or absence of an abnormality in the transport of the substrate **11A** at the C position by processing the image obtained by photographing the substrate **11A** that has reached the C position **P3A**. In this case, from the perspective of minimizing the burden on the abnormality detection unit **7** through image processing, etc., the detection of an abnormality is only necessary when the substrate **11A** reaches the C position **P3A**.

[0035] According to this embodiment, the timing at which the substrate **11A** reaches the C position **P3A** can be identified using the A sensor **12A**, the B sensor **13A** and the timekeeping section **14**. In other words, if the time required for the transport section **3A** to transport the substrate **11A** from the B position **P2A** to the C position **P3A** is known, the elapsed time can be measured by the timekeeping section **14** when the A sensor **12A** and the B sensor **13A** detect the presence of the substrate **11A** in this order, and it can be judged that the substrate **11A** has reached the C position **P3A** when the elapsed time reaches the above-mentioned time required. Also, when the sensors **A12A** and **B13A** detect the presence of the substrate **11A** in the reverse order, it is possible for the abnormality detection unit **7** not to detect the presence or absence of an abnormality (preferably, without photographing using the abnormality detection unit **7**, and even more preferably, without photographing using the abnormality detection unit **7** and measuring the elapsed time using the timekeeping section **14**). Therefore, the abnormality detection unit **7** can detect the presence or absence of an abnormality in the transport of the substrate **11A** only when the substrate **11A** has reached the C position **P3A**, based on the elapsed time measured by the timekeeping section **14** and the image obtained by the photographing. In addition, the A sensor **12A** and the B sensor **13A** can be configured with simple sensors such as laser sensors, for example, and the burden of data processing can be reduced. Therefore, it is possible to realize a substrate transport system **1** that can reduce the burden of data processing when detecting, by photographing, the presence or absence of abnormalities in the transport of the substrate **11A**.

[0036] The abnormality detection unit **7** performs the photographing using the imaging section **7aA** at a timing determined based on the elapsed time measured by the timekeeping section **14**.

According to the above configuration, by specifying the timing at which the substrate **11A** reaches the C position **P3A** using the A sensors **12A**, the B sensor **13A**, and the timekeeping section **14**, and then performing the photographing using the abnormality detection unit (imaging section **7A**), it is possible to avoid performing photographing at unnecessary times when detecting the presence or absence of an abnormality in the transport of the substrate **11A**. This allows to reduce the burden of photographing on the abnormality detection unit **7**, and further reduce the burden on data processing thereof. The image photographed by the abnormality detection unit **7** can be a still image in this embodiment, but it is not limited to this and can also be a video image. If the image captured by the abnormality detection unit **7** is a video image, the abnormality detection unit **7** may

be configured to start and stop photographing at timings determined based on the elapsed time measured by the timekeeping section **14**. The imaging section **7aA** is configured with a camera, etc., such as a CCD (Charge Coupled Device) image sensor.

[0037] The abnormality detection unit **7** detects the presence or absence of an abnormality in the transport of the substrate **11A**, using the judgment section **7b**, based on the image obtained by the imaging section **7aA**. In this embodiment, as the presence or absence of an abnormality in the transport of the substrate **11A**, the presence or absence of an abnormality in the clearance **C** between the main surface of the substrate **11A** (in this embodiment, the top surface of the disk-shaped substrate **11A**) and the system component **6A** is detected. The judgment section **7b** and the timekeeping section **14** are configured with a processing device **8** (computer, etc.) that can process the images obtained by the imaging section **7aA**.

[0038] The substrate transport system **1** has the gripper **4aA** as a system component **6A** and the pallet **4bA**, and the gripper **4aA** holds the substrate **11A** between a receiving part **4dA** of the pallet **4bA** and the gripper **4aA** itself by moving forward with respect to the pallet **4bA**, and releases the substrate **11A** by moving backward with respect to the pallet **4bA**. In addition to the gripper **4aA** and the pallet **4bA**, the transport pallet **4A** has a plurality of other grippers **4cA**. The number of other grippers **4cA** provided on the transport pallet **4A** is two in this embodiment, but this is not limited to this number. The transport pallet **4A** may be configured without the other grippers **4cA**. The gripper **4aA** and the other grippers **4cA** can be formed using fluororesin such as polytetrafluoroethylene as a material. The receiving part **4dA** for the gripper **4aA** and the other receiving parts **4dA** for the other grippers **4cA** can be formed using materials such as engineering plastics such as PEEK (poly ether ether ketone).

[0039] Transport section **3A** performs the first transport (see the white arrow pointing to the right in FIG. **1**) that transports the substrate **11A** toward the system component **6A** in the forward direction so that the substrate **11A** reaches the A position **P1A**, the B position **P2A**, and the C position **P3A** in this order; and the second transport (see the thick arrow pointing to the left in FIG. **1**) that transports the substrate **11A** away from the system component **6A** in the backward direction so that the substrate **11A** reaches the C position **P3A**, the B position **P2A**, and the A position **P1A** in this order.

[0040] The transport section **3A** is configured with a robot arm with a placement section **3aA** on which the substrate **11A** is placed. The substrate **11A** which has been placed on the placement section **3A** is placed on the pallet **4bA** of the transport pallet **4A** by the first transport performed by the transport section **3A**, and is held between the gripper **4aA** and the receiving part **4dA** of the pallet **4bA**. The substrate **11A** is transported into the processing section **5A** by the transport pallet **4A** in this state, and after the prescribed processing, the substrate **11A** is returned to its original position by the transport pallet **4A** (see the double-headed thick arrow in FIG. **1**). Furthermore, the substrate **11A** is released by the retraction of the gripper **4A**, and is transported from the transport pallet **4A** by the second transport performed by the transport section **3A**.

[0041] As illustrated in FIG. **5**, the substrate transport system **1** has a notification section **9** that makes a notification when the abnormality detection unit **7** detects the presence of an abnormality in the transport of the substrate **11A**. According to the above configuration, by notifying the system administrator **10** of the abnormality through a notification, it is possible to call their attention or prompt them to take measures such as inspecting the substrate transport system **1** or stopping its operation. The notification section **9** can be configured with a monitor **9a** that provides notification by displaying information on the screen, a warning light **9b** that provides notification by light, or a buzzer (not illustrated) that provides notification by sound, but this is not limited to these.

[0042] As illustrated in FIG. **5**, the substrate transport system **1** has: a second transport section **3B** that transports a second substrate **11B** so that the second substrate **11B** reaches second A position **P1B**, second B position **P2B**, second C position **P3B**, the second B position **P2B**, and the second A position **P1B** in this order, a second A sensor **12B** that detects presence of the second substrate **11B**

that has reached the second A position P1B, and a second B sensor 13B that detects presence of the second substrate 11B that has reached the second B position P2B, wherein the timekeeping section 14 measures second elapsed time when the second A sensor 12B and the second B sensor 13B detect the presence of the second substrate 11B in this order, the abnormality detection unit 7 performs second photographing of the second substrate 11B that has reached the second C position P3B, and detects presence or absence of an abnormality in transport of the second substrate 11B based on the second elapsed time measured by the timekeeping section 14 and an image obtained by the second photographing, and the notification section 9 makes a notification when the abnormality detection unit 7 detects the presence of an abnormality in transport of the second substrate 11B at the second C position P3B. According to the above configuration, it is possible to use the common judgment section 7b, timekeeping section 14 and notification section 9 to detect the presence or absence of an abnormality in the transport of the (first) substrate 11A at the (first) C position P3A and to notify of this, as well as to detect the presence or absence of an abnormality in the transport of the (second) substrate 11B at the (second) C position P3B and to notify of this, therefore, efficient operation of the entire system can be achieved. It will be noted that the substrate transport system 1 is not limited to the above configuration that uses a common judgment section 7b and timekeeping section 14. For example, the system 1 may be configured to have a first judgment section and a first timekeeping section (not illustrated) for detecting the presence or absence of an abnormality in the transport of the first substrate 11A at the (first) C position P3A, and a second judgment section and a second timekeeping section (not illustrated) for detecting the presence or absence of an abnormality in the transport of the second substrate 11B at the (second) C position P3B.

[0043] The abnormality detection unit 7 performs the photographing using the second imaging section 7aB at a timing determined based on the second elapsed time measured by the timekeeping section 14. The abnormality detection unit 7 detects the presence or absence of an abnormality in the transport of the second substrate 11B, using the judgment section 7b, based on the image obtained by the second imaging section 7aB. As the presence or absence of an abnormality in the transport of the second substrate 11B, the presence or absence of an abnormality in the clearance C between the main surface of the second substrate 11B and the second system component 6B is detected. Although the embodiment of the present disclosure has been

[0044] described above, the present disclosure is not limited to the above-described embodiment, and the above-described embodiment can be modified in various ways without departing from the gist of the present disclosure.

Claims

1. A substrate transport system having: a transport section that transports a substrate so that the substrate reaches A position, B position, C position, the B position, and the A position in this order, a A sensor that detects presence of the substrate that has reached the A position, a B sensor that detects presence of the substrate that has reached the B position, a timekeeping section that measures elapsed time when the A sensor and the B sensor detect the presence of the substrate in this order, and an abnormality detection unit that photographs the substrate that has reached the C position, and detects presence or absence of an abnormality in transport of the substrate based on the elapsed time measured by the timekeeping section and an image obtained by the photographing.
2. The substrate transport system as described in claim 1, wherein the abnormality detection unit performs the photographing at a timing determined based on the elapsed time measured by the timekeeping section.
3. The substrate transport system as described in claim 1, having a notification section that makes a notification when the abnormality detection unit detects the presence of an abnormality in transport of the substrate.

4. The substrate transport system as described in claim 3, having: a second transport section that transports a second substrate so that the second substrate reaches second A position, second B position, second C position, the second B position, and the second A position in this order, a second A sensor that detects presence of the second substrate that has reached the second A position, and a second B sensor that detects presence of the second substrate that has reached the second B position, wherein the timekeeping section measures second elapsed time when the second A sensor and the second B sensor detect the presence of the second substrate in this order, the abnormality detection unit performs second photographing of the second substrate that has reached the second C position, and detects presence or absence of an abnormality in transport of the second substrate based on the second elapsed time measured by the timekeeping section and an image obtained by the second photographing, and the notification section makes a notification when the abnormality detection unit detects the presence of an abnormality in transport of the second substrate.
